

VOLKSWAGEN

AKTIENGESELLSCHAFT

Quality management agreement between the companies of the VOLKSWAGEN GROUP and small-batch production suppliers



The *micro* Formula Q-Concrete contains contractually agreed specifications for companies in the VOLKSWAGEN GROUP in order to assure the quality of parts and processes in the procurement and supply chain.

October 2008

1st edition

1st edition – October 2008

The German-language version of the *micro* Formula Q-Concrete is binding.
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Issued by: Volkswagen AG
Group Quality Control
Brieffach 1466/0 38448 Wolfsburg, Germany

A note to suppliers

Dear Sir/Madam,

Increasing demands of customers, global competition and pressure costs require products for series launch to be mature and their production processes robust. It is essential that we work together in facing up to this challenge to ensure the market success of our products.

The booklet you are holding represents the first edition of the *micro* Formula Q Concrete. It lists essential small-batch production requirements regarding quality which you, as a supplier of production materials, are required to meet. The *micro* Formula Q Concrete and the other applicable documents form the basis for requests for quotations and must hence be taken into account when submitting a quotation at all times.

To ensure that our collaboration is successful, it is essential that the requirements described in this contract are adhered to bindingly across the supply chain while open communication, costs and deadlines are maintained.

As our potential supplier, when you submit a quotation, you simultaneously acknowledge your awareness of the requirements described in the *micro* Formula Q-Concrete and your agreement to comply with them without reservation and also to ensure they are carried out within your supply chain.

The *micro* Formula Q-Concrete plus future updated versions of this can be found on the Internet at: www.vwgroupsupply.com

Wolfsburg, October 2008



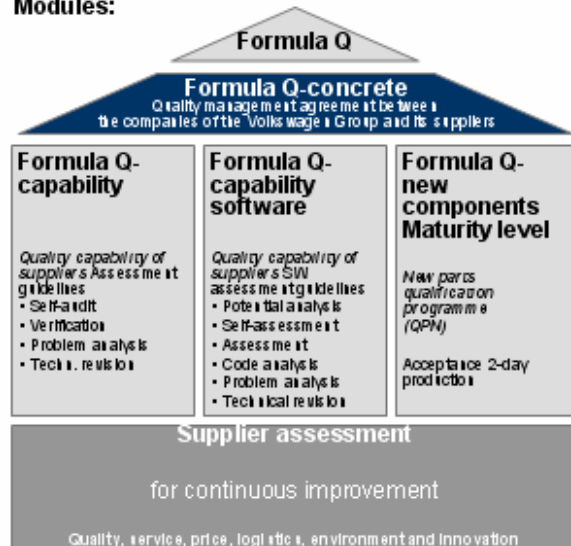
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Quality Management Agreements Purchased Parts

Modules:



"Philosophy"

Higher-level agreement as an integral part of the contract

Assessment systems and supporting processes

Other applicable documents in their relevant editions

These documents are binding for parts liable for documentation. For parts for which documentation is not obligatory, these documents apply as a basis for additional individual agreements.

Group quality documents at www.vwgroupsupply.com:

- Q-Capability Formula
- Q-Capability Formula software
- *micro* Formula Q New Parts Maturity
- TF/Parts Dispatch Control form
- Manual for Genuine Parts Suppliers
- Basic Corporate Software Requirements
- Group Guidelines: Wiring Harness Product Audits
- Group Packaging Guidelines (Engine Units and Vehicle)
- Supplier's Handbook for Prototypes
- Q Certification for the Pre-Series Phase
- QA Guidelines: Logistics and CKD
- Framework Agreement on Concept Responsibility
- Rules regarding the passing through costs of field and 0km complaints caused by the supplier
- Target Agreement on Quality Capability

Group Standard Directory at www.vwgroupsupply.com (FE Online Standards Texts):

- High-strength fasteners, screws and nuts (VW 60250)
- Machine capability test for measurable features (VW 10130)
- Process capability test for measurable features (VW 10131)
- Testing process suitability – Observance of measurement reliability in testing processes (VW 10119)
- General requirements for service provision as part of component development (VW 99000)

Publications by the German Automobile Association (VDA) and automotive standards at www.vda-gmc.de:

- VDA publications “Quality Management in the Automotive Industry” and “Mutual Quality Management within the Supply Chain” (series of documents by the Automobile Industry Association)
- ISO/TS 16949, alternatively VDA 6.1

Group sustainability documents under “Environment and Social Matters” at www.vwgroupsupply.com

In addition to the above, relevant technical supplier regulations and VW standards such as minimum shelf lives for components and legal stipulations, ordinances etc. apply for the product in question in each case.

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1. Queries and offers

1. Queries and offers

Definition of small-batch production

Small-batch production quantities: < 15,000 per duration of contract until EOP (including special editions).

1.1 Offer prerequisites

Before making an offer, the supplier must enter all supplier details concerning production and development sites at www.vwgroupsupply.com. If any information is missing, the supplier will be omitted from the group of bidders.

1.2 Query documentation

The contractor is obliged to check all requirements contained in query documents for completeness, freedom from contradictions, feasibility and technology versions (see also VW 99000, Chap. 2.5) and inform the customer in writing of any deviations (escalation).

Any necessary changes or additions such as paint quality, residual dirt values, airbag safety equipment etc. are to be cleared with the departments responsible before the offer is made (e.g. for product and process requirements) and documented in writing.

For certain product groups such as high-strength screws and cable harnesses, special Volkswagen Group-specific requirements must be met. These are published on the B2B platform and included in the relevant queries.

1.3 The supplier concept

The companies of the VOLKSWAGEN GROUP define three categories for development levels:

- Design developer
- Series developer
- Series supplier

Depending on the development level assumed in each case and after consultation with the responsible departments at the recipient, the supplier draws up an offer containing at minimum the elements below. This is to be presented by the supplier to the quality assurance department responsible when requested (e.g. Purchased Parts Quality Assurance) prior to the contract award process.

- Explanation of design concept (e.g. geometry, materials, functions, software)
- Project organisation planned with responsible contact partners for the development and production site
- Demonstration of the planned production processes, factory layout and supply chains
- Explanation of a deadline schedule for a parts approval target forecast
- Details of testing and approval planning (manufacturing chain including recipient)
- Explanation of sub-supplier and change management
- Explanation of measures for attaining quality targets
- Statement of agreement on target costs, deadlines, unit quantities

1. Queries and offers

- Risk assessment for deadlines, costs and quality
- Stipulation of a cost bearer for cost-labile special measures
- Binding statement of feasibility based on requirements specification (see VDA Assurance of Maturity of New Parts, RF 2 and ISO/TS 16949 Chap. 7.2.2.2)
- Logistics and packaging concept

Any remaining deviations to the requirements must be reported in writing to the contact person in the Quality Assurance department at the relevant company of the VOLKSWAGEN GROUP and these escalated where applicable (gap analysis).

1.4 Quality framework agreement

Companies of the VOLKSWAGEN GROUP require a zero defect strategy in all cases. In addition to the supply contract regulations on defects and other liability claims, the supplier must agree with the responsible Quality Assurance department in writing to a quality improvement programme in the case of any defects occurring. If no written agreement exists, the supplier is obliged to an annual halving of the defect quota.

The supplier is answerable to the companies of the VOLKSWAGEN GROUP for the quality of the component/module/system supplied and controls and coordinates sub-suppliers in the production and supply chain. The supplier ensures, using relevant contractual regulations, that documentation valid for the relationship between the Volkswagen Group Company and the supplier is likewise observed in dealings with sub-suppliers in the production and supply chain.

If suppliers or sub-suppliers, are defined within the production and supply chain for a component/module/system by a Group company i.e. set parts suppliers, the set part supplier must conclude a quality framework agreement with its direct customer (i.e. the receiving supplier in the production chain). In this relationship too, the 1st tier supplier is fully answerable to the companies of the VOLKSWAGEN GROUP for the quality supplied.

However, 1st tier suppliers of components/modules/systems in quantities ≤ 500 units over the contract validity period are an exception.

With these components/modules/systems, the supplier must stipulate before the final contract is awarded whether the components of the module/system provided by the designated part supplier are to be inspected first-hand by a company of the VOLKSWAGEN GROUP or by the 1st tier supplier itself.

2. Quality criteria for being awarded the contract

2. Quality criteria for being awarded the contract

It is absolutely essential that all suppliers of safety-relevant parts and parts liable for documentation received a quality capability rating of A or B in the supplier appraisal.

The quality performance of suppliers that already supply the VOLKSWAGEN GROUP will be assessed too.

If no rating is available, a process assessment is required for the relevant production site before the contract can be awarded in the CSC process.

An unsuitable (C) rating in one of the two criteria leads to exclusion from the contract award process.

For other parts, a binding improvement programme with the goal of achieving at least a B rating may have to be agreed before series supply can commence.

The supplier is obliged to inform the certification company of a C classification (business on hold). The receiving company in the VOLKSWAGEN GROUP must be involved in the subsequent procedure between the certification company and the supplier.

2.1 Elements of supplier appraisal

- Proof of an operational quality management system by a certificate from an IATF-registered certifying company as per ISO/TS 16949 or VDA 6.1. Without this certificate, the quality capability rating A cannot be awarded.
- Process suitability is assessed according to Formula Q capability and/or Formula Q capability software. Results are given an A, B or C rating.
- Quality performance is assessed in accordance with experience from previous projects as well as supply and field quality. Escalation to C rating is done using the Critical Series Suppliers programme.
- Any product or project-specific risk assessments have to be done with the involvement of experts from the Purchased Parts/Procurement Quality Assurance Departments and Technical Development Department prior to the series being awarded. An improvement programme is devised from these where applicable with essential measures for the actual project.

2.2 Target agreement on quality capability

a) Parts liable for documentation

To ensure quality of components/modules/systems, the companies of the VOLKSWAGEN GROUP requires the production of the components/modules/systems to be carried out by the supplier at a production site with an A rating.

If the quality capability of the supplier's production site has been rated B at the time the contract is awarded, the supplier is obliged to qualify the relevant site so that it obtains an A rating. If this requirement is not met, escalation takes place. Details are regulated in the nomination letter and target agreement on quality capability.

b) All other parts

Where the quality capability of the supplier's production site is not classified at the time the contract is awarded, the supplier is obliged to qualify the relevant production site so that quality capability is attained no later than the start of series production.

2. Quality criteria for being awarded the contract

2.3 Framework agreement on design responsibility (KVV)

To bindingly define responsibilities between the commissioning company of the VOLKSWAGEN GROUP and the supplier early during development of the components/modules/systems, a design responsibility agreement (KVV) is used. Unambiguously regulating responsibility for the component design gives suppliers greater influence on joint products with Group companies. The KVV comprises a KV Framework Agreement, which is concluded with the supplier each time a new contract is awarded for components/modules/systems plus an accompanying KV Quota Agreement sheet – KV Quota for short – concluded for each component.

The KV Quota always depends on the development level for the component. The following rule of thumb applies: the more design specifications are made by the Volkswagen Group, the smaller is the KV quota. The fewer design specifications are made by the Volkswagen Group, the greater is the KV quota.

The KVV is essentially determined by the length of the business relationship between the Group company and the supplier.

The component-specific KV Quota agreed is used to process unsatisfactory deliveries by the supplier whenever an analysis of the defective components/modules/ systems indicates the defect is due to its design or development.

The KVV or KV Quota is not used for the handling of campaigns, in particular with recalls, series damage or product liability cases. The KV Quota is also not used when the result of the analysis of the defective components/modules/systems ends in "Trouble Not Found" (kFf/ tnf).

3. Cooperation with suppliers during the product development process

3. Cooperation with suppliers during the product development process

3.1 Qualification program for new parts for small-batch production (*micro QPN*)/ maturity assurance/purchased parts management

The supplier is obliged to apply this qualification program (QPN) across its entire supply chain in order to create the preconditions for a robust production process for new parts (see also Formula Q for New Parts).

During the product planning, process planning and tool making phases, project progress will be determined for the delivery scope. In the process, individual parts having an identical or similar manufacturing process will be grouped into part families and their product maturity will be assessed. The companies of the VOLKSWAGEN GROUP reserve the right to assess the current project status of projects at any time by conducting a review at the supplier and/or sub-supplier.

Fixed defined rules including the initiation of what are known as “round table meetings” involve both suppliers (or sub-suppliers) of maturity-critical delivery scopes and the customer’s internal representatives in the product development process at an early stage (see also VDA Maturity Level for New Parts).

Important milestones:

- Design Freeze ((DF); Planning Approval)
- Procurement approval BF
- Official project discussions: Ascertainment of maturity at agreed times (milestones) in accordance with Maturity for New Parts (see also VDA Maturity for New Parts and Formula Q for New Parts Integral).
- Kick-off meetings: Initial meetings for delivery scope with risk potential across the entire process chain.
- Pre-series approval vehicles (VFF): Initial production using tools intended for series production with documentation of status to enable risks to be detected at an early stage.
- Production Test Series (PVS): Proof of manufacturing feasibility for the planned products using series tools and series systems at the ultimate locations.
- Zero series (0 series): Proof of series suitability using the final system status. Following approval of parts status, the series start-up (process validation and 2-day production, “2 DP”) is validated. For smaller quantities, instead of carrying out 2 DP, it can be agreed that the supplier provide proof of manufacturing feasibility over the duration of the production period based on a sample size of 1.5 times the weekly volume using materials at hand.
- Parts provision deadline: Suppliers’ purchased parts must be delivered by the parts provision deadline for the relevant pre-series (VFF, PVS, zero series).
- Start of Production (SOP): Start-up curve is confirmed by supplier.

4. Ongoing quality measures during series production

4. Ongoing quality measures during series production

4.1 Ongoing validation of process capability

Process capability is a measure of the quality of a process based on the specification of the products to be produced during that process.

Where it is not possible to verify process capability (e.g. manual work or rectification work), critical feature specifications must be agreed for the testing of 100% of parts by the supplier. Lot sizes are to be agreed based on the shelf life of parts and availability of the supplier's own, appropriate warehouse space.

A validation concept for critical attributive features (those which may lead to A and B defects) must be presented and agreed with the recipient. This may be realised, for instance, using multiple validation e.g. through poka yoke, labour organisation measures, measurements on the production line, audits, 100% tests and so on for the zero-defect acceptance criterion as per ISO/TS 16949, Chap. 7.1.2.

For each process or product risk identified (e.g. by FMEA), it is essential that at least one method or tool be applied in order to attain quality goals in keeping with the latest technology status as described in VDA Volumes 4 and 14.

4.2 Product safety, product liability

When using purchased parts, the companies of the VOLKSWAGEN GROUP bear the final manufacturing responsibility and overall responsibility for the final product – the vehicle.

Primary manufacturing responsibility for the purchased parts installed in the final product lies with the supplier or, where applicable, the sub-supplier. Suppliers are therefore obliged to take all organisationally and technically feasible and reasonable steps to increase the product safety of their parts and the parts of their sub-suppliers and minimise the risks of product liability.

The supplier shall ensure and oblige its sub-suppliers to ensure that:

- a distinct awareness of quality exists throughout the whole company
- the necessary product safety is guaranteed during development of components
- particular attention is paid to product safety during quality planning
- quality capability of manufacturing processes is guaranteed and verified
- the likelihood of defective products occurring is minimised by ongoing quality assurance measures during series production
- timely discovery and management of defective products during ongoing production is guaranteed using appropriate measures (minimisation of costs and/or waste of added value)
- quality data and legally-required verification tests are documented in detail to prove products have been manufactured in accordance with statutory requirements and safety standards
- a material tracking system is used so that the effects of any defects which occur can be restricted as the need arises
- responsible staff receive detailed instruction and training on product safety and product liability
- systems comparable with Formula Q Concrete, etc. which satisfy recipient requirements are used by all sub-suppliers

4. Ongoing quality measures during series production

- a product safety officer (PSB) is nominated for each step in the supply chain. The PSB for the 1st tier must be entered in the suppliers' database (LDB). This must be kept up to date at all times.
- components with limited shelf life meet specific marking requirements, especially in accordance with the Supplier's Manual for Genuine Parts.

4.3 Internal audits

In order to assess and improve the quality capability of business sectors, the supplier is obliged to carry out internal audits on a regular basis.

In addition to this, the supplier must carry out at least 1 D/TLD self-audit as per Formula Q Capability each year for delivery scope items designated as D parts.

The results of D/TLD audits must be archived for at least 15 years.

The companies of the VOLKSWAGEN GROUP reserve the right to perform process and product audits at any time.

4.4 Control cycles

4.4.1 Complaint handling

The companies in the VOLKSWAGEN GROUP maintain a comprehensive defect management system. The supplier is obliged to systematically rectify any defects that occur as quickly as possible within the framework of existing regulation and demonstrate the sustainability of its measures to the Quality Assurance department responsible.

The responsible quality assurance departments, logistics centres and all other partners in the supply chain must be informed by the supplier of any defects.

An 8D report must always be commenced for all defects. The complaint-handling process may not be regarded as finished until this report has been concluded.

In case of A and B defects and on request, the supplier is obliged to rectify these using immediate measures (see also Formula Q Capability, Chapter 5.3). This means at minimum:

- Immediate separation/rectification of stock on site at all recipients
- Use of a 100% firewall to prevent any other unnoticed defects occurring.

If the supplier cannot implement this using its own staff, the supplier is obliged to engage third parties for the handling.

With A defects and on request, the supplier is obliged to dispatch within 24 hours a representative authorised to make a decision to the Quality Assurance department responsible at the receiving company of the VOLKSWAGEN GROUP.

4.4.2 Early warning system

Complaint processing provides the Group company and its suppliers with important early warning information on new and hitherto unknown problems with products.

Indications of problems in the supplier's plant:

Where significant deviations in defect key figures are ascertained for the supplier's products (e.g. scrapping and rectification work), the supplier must inform the quality assurance department responsible about this without delay.

This also applies where defects are detected in similar products of competitors.

4. Ongoing quality measures during series production

Indications of problems at the receiving company of the VOLKSWAGEN GROUP (Production hall incidents):

The supplier is obliged to process any information provided by the receiving company of the VOLKSWAGEN GROUP immediately so that defective parts are prevented from reaching the final customer. Quality information regarding incidents must be taken notice of and assessed regularly and promptly.

Indications of problems in the field:

For the early identification of defects in the field, the supplier will provide support for monitoring and evaluating relevant early warning systems (hotlines, retail task force etc.) e.g. using resident engineers. With new product launches or with 100% obligation to report damaged parts in the market under observation, the supplier is obliged to accompany the defect rectification process on site.

4.4.3 Obligation to own field monitoring with mandatory reporting

As part of mandatory product monitoring, the supplier is also obliged to carry out independent market monitoring of the products it launches directly onto the market and inform the receiving company of the VOLKSWAGEN GROUP of any transferable knowledge.

4.5 Continuous improvement process (KVP)

The supplier is obliged to provide evidence of a continuous improvement process in accordance with ISO/TS 16949 Chapter 8.5. In order to do this, the supplier will use suitable measures to reduce internal scrapping and rectification work rates. This information must be provided to the VOLKSWAGEN GROUP on request.

4.6 Change management

The supplier is obliged to inform the receiving company of the VOLKSWAGEN GROUP of any changes to its process chain (location, product, process) before implementing these and then to obtain agreement from the responsible Quality Assurance department. The need for new sampling must be coordinated with the responsible Quality Assurance department at the receiving company of the VOLKSWAGEN GROUP. If the supplier fails to observe this, the companies of the VOLKSWAGEN GROUP reserve the right to initiate the "Critical Series Suppliers" escalation programme and, where applicable, issue a C classification (business on hold, see Chap. 2). Any resulting additional costs for the new approval process shall be borne by the supplier.

4.7 Re-qualification

For quality assurance purposes, the supplier is obliged to perform re-qualification of the delivery scope on a regular basis in accordance with ISO/TS 16949 (Chapter 8.2.4.1) and VDA Robust Production Processes (Chapter 5.3.4). Any deviations in the contents of this re-qualification must be agreed between supplier and recipient. The companies of the VOLKSWAGEN GROUP require re-qualification to be performed in full at least every five years. Deviations to this regulation can be agreed in writing with the companies of the VOLKSWAGEN GROUP.

4.8 Lessons learned

4. Ongoing quality measures during series production

Experience gained from previous and current projects such as malfunctions in the field, production hall incidents, project handling etc. is to be used as “lessons learned” for new projects and developments and to demonstrate a measurable reduction in defects during new product launches based on these previous indicators.

4.9 Cost bearer for additional costs caused by deviations

Additional expenses caused by field malfunctions and production hall incidents shall be passed through to the causing party by the companies of the VOLKSWAGEN GROUP.

A technical factor (TF) shall be specified between the companies of the VOLKSWAGEN GROUP and their suppliers for determining supplier responsibility for field malfunctions in accordance with the relevant document of the Purchased Parts Quality Assurance department at the respective Group company (e.g. rules regarding the passing through of costs of field and 0km complaints caused by the supplier). Causes for malfunctions are principally grouped into process and design defects.

Technical factors are defined for specified periods of time.

4.10 Technical Audit of Suppliers (TRL)

The companies of the VOLKSWAGEN GROUP always pursue the following goals for the TRL (see Formula Q-Capability) worldwide:

- Assuring that a part or component satisfies specified requirements
- Testing on-site series production and all other validation activities
- Checking effectiveness of corrective measures and verification of agreed quality management standards

The companies of the VOLKSWAGEN GROUP reserve the right to perform a TRL (Technical Audit of Supplier) at any time. The TRL shall be announced one working day before it is performed.

Assessment of the TRL is made using traffic light symbols. A red traffic light leads to classification under Level 2 in the Critical Series Suppliers programme and obliges the supplier at short notice to quality meetings with the quality assurance department responsible for the TRL at the receiving company of the VOLKSWAGEN GROUP together with the top management of the supplier, during which a decision is made about a C classification (business on hold; Chap. 2).

4.11 Critical Series Suppliers programme

In case of serious discrepancies in quality requirements such as supplier performance, sampling or a red TRL assessment, the responsible Quality Assurance department at the receiving company of the VOLKSWAGEN GROUP escalates the supplier in the “Critical Series Suppliers” programme.

The programme comprises four defined levels:

Level 0: The supplier has problems

Level 1: The supplier is not successful in solving the problems

Level 2: The supplier requires help from a third party to ensure delivery capability, C warning

4. Ongoing quality measures during series production

Level 3: The supplier is not suitable for Group quality, (business on hold; Chap. 2)

Classifications are usually subject to a time limit. If the measures implemented are proven to be sustainable, de-escalation is performed.

In individual, well-founded cases, Group Quality Assurance reserves the right to permit direct classification in Level 3.

Terms and Abbreviations

Terms and Abbreviations

0km defect	Defect which becomes apparent during assembly of parts on the assembly line or delivery.
0 series	Zero series is the time the first new vehicles are assembled on the production line under series production conditions. At this time, an initial sampling grade of 1 is required for purchased parts.
1st tier supplier	A direct supplier to a receiving plant of a company of the VOLKSWAGEN GROUP
2-day production (2 DP)	To ensure the quality and quantity of the production launch, process parameters are verified during what is known as " 2-day production "
A, B, C defects	For classification of defects with regards to severity, the following classes exist: A1 Safety risk, tow-in A Not acceptable, will doubtlessly lead to customer complaint B Objectionable, disturbing, customer complaints expected, deviation from specifications, production stops at receiving plant are possible C Need for improvement, customer complaints and disturbances during production to be expected if defect occurs frequently
Attributive features	Attributive features are features for which only the number of defects is determined: - Proportion of defective units (e.g. 1.8% of bolts defective) - Average number of defects per unit (e.g. 3 paint defects per vehicle)
B2B	Business to Business (e-commerce platform on Internet for communication between suppliers and companies in the VOLKSWAGEN GROUP) Internet: www.vwgroupsupply.com Phone number in Germany: 0800 - 193 30 99 International phone number: +49 - 5361 - 93 30 99 E-mail: supplierintegration@vwgroupsupply.com
Special features	Critical and function-relevant product, process and test features are to be specified using the "System FMEA Product" in cross-divisional teams. Other special features may arise from the subsequent "System FMEA Process". Besides statutory, safety-relevant, design and process-oriented aspects, these also include key customer-oriented aspects.
Business on hold	A supplier production site is blocked for further orders → C classification

Terms and Abbreviations

Design Freeze	Conclusion of detailed design and planning approval
D feature	Product feature requiring documentation (see also ISO/TS 16949 Chap. 7.3.2.3)
D part	Component with a feature for which documentation is mandatory (see also ISO/TS 16949 Chap. 7.3.2.3)
D/TLD	Documentation Mandatory/Technical Guideline Documentation (see Formula Q Capability, Chap. 8, p. 1)
Firewall	Safety component which, based on a defined rule, prevents unsatisfactory parts slipping through the production line.
FMEA	F ailure M ode and E ffect A nalysis; analysis of potential downtime and its consequences.
Gap analysis	A gap analysis serves to identify strategic and operational gaps by comparing target specifications and probable developments in the sector under analysis.
IATF	I nternational A utomotive T ask F orce (see ISO/TS)
ISO/TS	I nternational S tandard O rganisation/ T echnical S pecification
Kick-off meetings	Initial discussion of fundamental issues
KVP	K ontinuierlicher V erbesserungs p rozess (Continuous improvement process)
KVV	K onzept v erantwortungs v ereinbarung (Design Responsibility Agreement) (see other applicable documents)
LDB	L ieferant e n d aten b ank (Suppliers' Database)
OEM	O riginal E quipment M anufacturer (Automobile manufacturer)
Poka Yoke	System for preventing accidental malprocessing
Critical Series Suppliers programme	Escalation process for suppliers exhibiting inadequate performance in the production hall (Okm) or field. The process can lead to the supplier being temporarily blocked for new contracts (business on hold). (See other applicable documents)
PSB	P rodukt s icherheits b eauftragter (Product Safety Officer)

Terms and Abbreviations

PVS	ProduktionVersuchs-Serie (Production Test Series) is used for final refinement in assembly of individual parts. Assembly must take place using representative series production processes with parts directly from the tool. A rating of at least 3 is required for purchased parts.
Q	Quality
QPN	Qualification Programme for New Parts (see Formula Q New Parts Integral)
QA	Quality Assurance
RPP	Robust Production Process (see VDA Product Manufacturing and Delivery)
SOP	Start of Production (series launch)
Supply Chain	Supply Chain: 1st tier to nth tier
Task force	Group set up to solve a problem.
TF	The technical factor is worked out based on a mutually agreed technical analysis and determines the causing party's share of the damage cases examined. $TF = \frac{\text{Defects caused by supplier}}{\text{Examined defects}}$ (See rules regarding the passing through of costs of field and 0km complaints caused by the supplier)
TLD (catalogue)	List in which all characteristics liable for documentation are entered. (see ISO/TS 16949 Chap. 7.3.2.3)
TRL	Technische Revision Lieferant (Technical Audit of Supplier)
Trouble-not-found parts (tnf)	No defect could be determined: Defective parts for which the cause(s) of the defect have/has not yet been clarified.
VDA	Verband der Automobilindustrie e.V. (German Automotive Industry Association)
VW Standard	Technical standards in the companies of the VOLKSWAGEN GROUP (e.g. VW 10540 Manufacturer Codes for Vehicle Parts)

Terms and Abbreviations

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